AI201 Mini-Project

Philippine Medicinal Plant Leaf Classification: Classical Machine Learning vs. CNN Benchmark

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1. INTRODUCTION

The Philippines harbors an estimated 13,500 plant species, of which approximately 3,500 possess medicinal properties [2]. These species constitute the traditional healing systems practiced across the archipelago and serve as primary healthcare resources for rural communities with limited access to modern medical facilities [3]. Despite their significance, accurate identification of Philippine medicinal plants remains a persistent challenge. Many species are visually similar at the macroscopic level. Accurate and efficient plant identification is valuable for herbalists, pharmacognosy researchers, and rural health practitioners who rely on these species for traditional medicine.

Automated plant identification using leaf imagery provides a practical, scalable alternative to traditional methods. Leaves are ideal targets for classification, they are the most accessible plant organs, they encode distinct morphological traits, such as shape, venation, color, and texture, and they can be photographed without destructive sampling [4]. However, scaling automated classification to 40 visually diverse species introduces significant challenges. Species often share a similar gross leaf morphology while differing only in subtle venation density or surface texture (high inter-class similarity). Conversely, leaves from the same species can vary wildly due to age, viewing orientation, and lighting conditions (high intra-class variation).

The dominant trend in image classification favors deep convolutional neural networks (CNNs), which learns hierarchical feature representations directly from pixel data. On large-scale plant datasets, transfer learning using ImageNet architectures like VGG [5] and ResNet routinely yields accuracies above 90%. In fact, the specific dataset used in this study was previously benchmarked using VGG19, achieving an accuracy of 92.67% [1]. Despite their high performance, deploying these deep models in resource-constrained environments, such as low-power embedded devices, field-ready mobile apps, or offline settings, remains difficult due to their massive memory footprints, heavy computational costs, and opaque "black-box" nature [4].

2. OBJECTIVES OF THE STUDY

The project aims to apply traditional image processing methods and classical machine learning algorithms in classifying leaf images of Philippine medicinal plants. Specifically, it aimed to:

1. Extraction of Leaf features using traditional image processing methods, and using a Dimensionality Reduction to identify the important features.
2. Training of Support Vector Machine (SVM) and selecting the Top-N Features, to solve the curse of dimensionality.
3. Optimization of best performing classical model using Grid Search and compare its performance to previous Deep Learning Models.
4. Highlight the aim to show the viability of manual feature extraction and classical methods in the era of deep learning.

3. REVIEW OF RELATED LITERATURE

3.1 Leaf-Based Plant Identification

Automated plant identification from leaf imagery has been an active research area. Early works established that morphological descriptors, including leaf area, perimeter, circularity, eccentricity, and solidity, carry discriminative information to distinguish plant species under controlled imaging conditions [7]. Subsequent studies expanded the feature to include texture descriptors, demonstrating that Gray Level Co-occurrence Matrix (GLCM) statistics and Local Binary Pattern (LBP) histograms capture surface microstructure differences that morphological features alone cannot resolve [8].

Kadir et al. [9] demonstrated that combining shape, color, and texture features within a single feature vector yields higher classification accuracy than any individual modality, a finding that directly motivates the multi-modal feature approach adopted in this study.

More work has incorporated vein structure as an additional discriminative modality. Leaf venation patterns encode phylogenetic information that is largely independent of leaf size and color variation, making vein features particularly robust to intra-class variation caused by developmental stage and imaging conditions [10].

3.2 Classical Machine Learning for Image Classification

Support Vector Machines with radial basis function kernels have consistently demonstrated strong performance on plant classification tasks characterized by moderate dataset size and high-dimensional feature vectors. Islam et al. [6] reported accuracies exceeding 90% using Histogram of Oriented Gradients (HOG) and LBP features with SVM classifiers on standard leaf benchmark datasets. Saleem et al. [4] conducted a broad survey of plant species and disease classification methods, concluding that classical feature engineering pipelines remain competitive with deep learning

approaches when the image domain is constrained and the training set is moderate size.

Ensemble methods, particularly gradient-boosted decision trees, have also been applied to plant classification tasks with favorable results. XGBoost, which constructs an additive ensemble of shallow trees via gradient descent on a differentiable loss function [11], is well-suited to tabular feature vectors derived from image descriptors and has shown competitive performance relative to SVM on several fine-grained recognition benchmarks. K-Nearest Neighbors, while conceptually simple, provides a useful non-parametric baseline that makes no assumptions about the underlying class conditional feature distributions [7].

3.3 Deep Learning for Plant Classification

Convolutional neural networks have advanced the state of plant image classification, particularly when applied to large and diverse datasets. Mohanty et al. [5] demonstrated that fine-tuning ImageNet-pretrained VGG and AlexNet architectures yields strong performance on plant disease classification, with the CNN-based approach outperforming classical methods on datasets comprising hundreds of thousands of images. However, the same study noted that the performance advantage of deep networks diminishes when training data is limited, as the models become prone to overfitting in low data. The PhilMedic dataset used in this study was introduced and benchmarked by Custodio [1], who reported a classification accuracy of 92.67% using VGG19 as the primary classifier, establishing the principal deep learning reference point which the classical pipeline developed in this study is evaluated.

3.4 Dimensionality Reduction in Feature-Based Pipelines

Feature selection methods based on statistical tests, such as ANOVA F-score ranking, address this by retaining only the features that exhibit statistically significant class-discriminative variance, yielding a compact representation that concentrates the most informative signal for downstream classifiers [12]. Selecting the top-k features by ANOVA F-score prior to SVM training has been shown to improve both classification accuracy and computational efficiency on moderate-sized image datasets.

4. METHODOLOGY

4.1 Dataset

The dataset used in this study is the PhilMedic: Philippine Medicinal Plant Leaf Dataset. It contains 4,922 high-resolution leaf images of 40 Philippine medicinal plant species. Images were captured using a 48MP Android camera against a white background, producing PNG files with a resolution of 3000×4000 pixels and an average file size of 3.02 MB per image, resulting in a total size of 7.76 GB. The dataset is openly available on Mendeley Data website. The class distribution is approximately uniform across the 40 species, with each class containing between 100 and 140 images.

4.2 Data Preparation

First, raw images were compressed into WebP format and resized to 500 × 500 pixels, drastically reducing the total dataset storage footprint, see Figure 1, without losing feature-relevant image quality. Second, background removal was applied to each image, eliminating background pixels that would otherwise introduce noise into color and texture feature computations. The resulting images contain the leaf subject on a uniform background, ensuring that all extracted features reflect leaf-intrinsic properties rather than imaging artifacts.

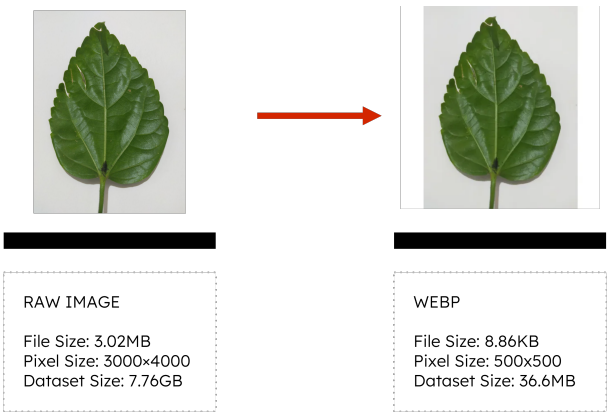


Figure 1: Raw leaf image (left) and background-removed WebP image (right) after preprocessing.

4.3 Feature Extraction

A 371-dimensional feature vector was constructed for each image by concatenating four descriptors: color features, texture features, shape features, and vein features.

Color Features. Color descriptors were computed in both the RGB and LAB color spaces. These include the green channel ratio relative to the red and blue channels, per-channel color moments comprising the mean, standard deviation, and skewness of each channel, color histograms capturing the distribution of intensity values, average saturation reflecting color vividness, and average brightness value.

Texture Features. Texture descriptors were derived from two complementary methods. Gray Level Co-occurrence Matrix (GLCM) statistics were computed to capture spatial co-occurrence patterns between pixel intensity pairs, yielding contrast, dissimilarity, homogeneity, correlation, and energy measures that quantify leaf surface smoothness and roughness. Local Binary Pattern (LBP) histograms were additionally computed to encode local micro-texture structure, capturing the distribution of regular and irregular surface patterns. Edge density, defined as the ratio of edge pixels to total pixels, and roughness, computed using the OpenCV implementation, were also included as supplementary texture descriptors.

Shape Features. Morphological shape descriptors were extracted using contour analysis via OpenCV, yielding area, perimeter, circularity, eccentricity, and solidity. Circularity is defined as $4\pi A/P^2$, where A denotes leaf area and P denotes perimeter, and measures the compactness of the leaf outline relative to a circle. Eccentricity, derived from the fitted ellipse, quantifies leaf elongation. Solidity is defined as the ratio of the contour area to the convex hull area and

captures the degree of leaf margin irregularity.

Vein Features. Vein structure was extracted using the Frangi vesselness filter, which enhances tubular structures in grayscale images and is well-suited to the detection of leaf vein networks. From the filtered vein map, the following descriptors were derived: vein density, defined as the proportion of vein pixels relative to the total leaf area; mean vein orientation, capturing the direction of the vein network; orientation standard deviation, measuring the angular spread of vein directions; and branching density, quantifying the frequency of vein bifurcation points per unit area.

The resulting feature matrix has dimensions $N \times 371$, where N denotes the number of images in the dataset. This matrix was exported as a CSV file and served as the input to the dimensionality reduction and classification stages.

4.4 Dimensionality Reduction

t-SNE and UMAP visualizations of the raw 371-dimensional feature space revealed overlapping plant classes. To address this, Top-N feature selection was applied using ANOVA F-scores.

The effect of k on cross-validation accuracy was evaluated systematically across a range from 10 to 371 features, with the optimal value of $k = 200$ determined via Grid Search as described in Section 4.5. Following Top-N feature selection, both t-SNE and UMAP projections produced markedly more compact and distinguishable clusters compared to the raw feature space, confirming that the selected subset concentrated the most discriminative signal.

4.5 Model Training

A Support Vector Machine (SVM) with a Radial Basis Function (RBF) kernel was selected as the primary classifier, trained on the Top-N selected feature vectors. Hyperparameter optimization was performed using Grid Search, jointly searching over the number of selected features k, the regularization parameter C, and the kernel bandwidth γ. Five-fold cross-validation accuracy was used as the optimization criterion. The optimal hyperparameter configuration identified by Grid Search was: $k = 200$ selected features, RBF kernel, $C = 100$, and $\gamma = 0.001$. This configuration achieved a best cross-validation accuracy of 97.38% and a final test accuracy of 98.06%. All models were implemented using scikit-learn. The complete software environment consisted of Python 3.12, scikit-learn, OpenCV, and UMAP-learn.

4.6 Evaluation Metrics

Model evaluation was performed on a held-out test set comprising 20% of the dataset. The primary metric was classification accuracy, computed as the proportion of correctly classified samples. A full classification report was additionally generated comprising precision, recall, and F1-score, each computed as weighted averages across the 40 species classes to account for minor class imbalance. A confusion matrix was generated to examine per-class classification behavior. ANOVA F-score analysis was also performed to identify the hand-crafted features that contributed most substantially to class separability, providing an interpretable account of which leaf properties are most discriminative for this dataset.

5. EXPERIMENTAL RESULTS

5.1 Effect of Feature Selection on Classification Performance

Table 1 summarizes SVM cross-validation accuracy across all evaluated values of k, the number of Top-N selected features. SVM accuracy increased sharply from 65.19% at $k = 10$ to approximately 97% for larger Top-N subsets, with diminishing returns observed beyond $k = 200$.

Table 1: SVM cross-validation accuracy (%) across Top-N feature subset sizes.

Number of Features (k)	CV Accuracy (%)
10	65.19
25	92.96
50	95.83
75	97.10
100	96.95
150	97.13
200	97.38
250	97.30
300	96.78
350	96.49
371	96.49

The Grid Search optimal configuration ($k = 200$, $C = 100$, $\gamma = 0.001$) achieved the highest cross-validation accuracy of 97.38%. Figure 2 illustrates the full accuracy curve across all evaluated values of k.

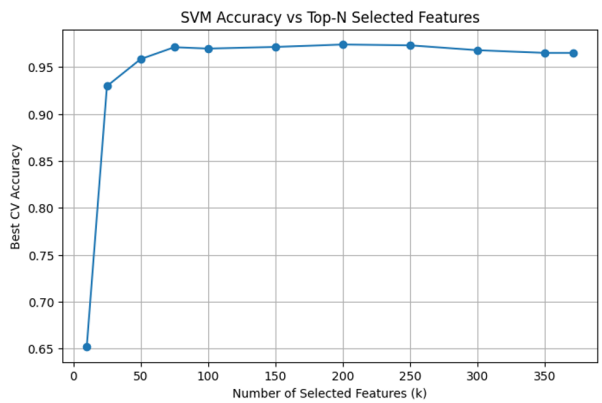


Figure 2: SVM cross-validation accuracy as a function of the number of Top-N selected features.

5.2 Comparison with Baseline Models

Table 2 presents the performance of the optimized classical SVM model alongside the deep learning baselines. The VGG19 Feature Extractor + SVM baseline uses VGG19 purely as a fixed feature extractor paired with an SVM classifier, while the VGG19 + CNN result reported by Custodio [1] represents the primary end-to-end deep learning reference point. All results in Table 2 are reported on the held-out test set comprising 20% of the dataset.

The optimized SVM with Top-N Feature Selection achieved 98.06% test accuracy, surpassing the published VGG19 + CNN result of 92.67% by a margin of 5.39 percentage points

Table 2: Performance comparison of the optimized SVM model against baselines.

Kind	Model	Accuracy (%)
Baseline	VGG19 + SVM	98.39
Classical	SVM + Top-N	98.06
Baseline	VGG19 + CNN [1]	92.67

and falling within 0.33 percentage points of the VGG19 Feature Extractor + SVM baseline.

5.3 Classification Report of the Best Model

Table 3 presents the full classification report for the best-performing SVM with Top-N Feature Selection evaluated on the held-out test set. The model achieves a weighted precision, recall, and F1-score of 0.98 across all 40 species classes, consistent with the overall accuracy reported in Table 2.

Table 3: Classification report for the best-performing SVM with Top-N Feature Selection on the 40-species test set.

	Precision	Recall	F1-Score
Accuracy			0.98
Macro avg	0.98	0.98	0.98
Weighted avg	0.98	0.98	0.98

5.4 Confusion Matrix

Figure 3 presents the confusion matrix for the SVM with Top-N Feature Selection evaluated on the 40-species test set. The matrix shows that the large majority of predictions are concentrated along the diagonal, indicating that the model correctly identifies most species.

5.5 Top-N Feature Importance

Figure 4 presents the top 20 most important features identified by ANOVA F-score analysis. Histogram-based color descriptors from the RGB, HSV, and LAB color spaces achieved the highest individual scores by a substantial margin, while morphological, texture, and vein features contributed lower baseline values to the top rankings.

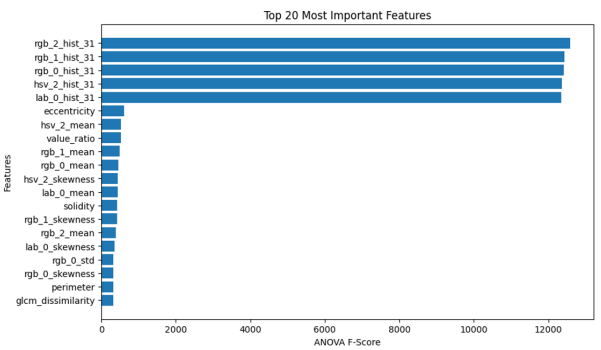


Figure 4: Top 20 most important features by ANOVA F-score.

6. ANALYSIS AND DISCUSSION OF RESULTS

6.1 Performance of Classic Machine Learning

The gap between the optimized SVM with Top-N Feature Selection (98.06%) and the VGG19 Feature Extractor + SVM baseline (98.39%) is narrow, at approximately 0.33 percentage points.This narrow gap suggests the discriminative bottleneck lies in feature representation quality, not classifier choice. The hand-crafted feature pipeline developed in this project approaches but does not fully close this gap, indicating that deep convolutional features, even when used non-adaptively, encode complementary information that the hand-crafted descriptors do not fully capture.

6.2 Role of Dimensionality Reduction

As qualitatively demonstrated by the t-SNE and UMAP projections in Figure 5, Top-N feature selection successfully addresses the manifold interleaving observed in the raw feature space. Following feature selection, both visualization methods showed markedly improved class separation, with the majority of species forming visually distinct clusters.

The success of Top-N feature selection in this setting can be attributed to the removal of redundant and weakly discriminative features from the 371-dimensional raw feature space. By retaining only the features with the highest ANOVA F-scores, the selected subset concentrates the most class-discriminative signal, enabling the RBF SVM to construct a more effective decision boundary. The sharp accuracy gains observed at small values of k confirm that a relatively compact subset of the hand-crafted features carries the majority of the discriminative information for this dataset.

6.3 Feature Importance

The prominence of color features reflects the meaningful chromatic variation across the 40 species, which includes species with dark green, light green, reddish, and variegated leaves. The LAB color space is perceptually uniform and separates luminance from chrominance, making LAB histograms more robust to illumination variation than RGB histograms while preserving the hue differences between species with distinct leaf pigmentation. The inclusion of HSV color descriptors further captures perceptual hue and saturation differences that complement the LAB representation.

Eccentricity ranked as the top shape feature, confirming that leaf elongation is a strong discriminative morphological descriptor for this dataset. LBP texture features appeared consistently across the top 20 features but at lower individual ANOVA F-scores, suggesting that texture contributes cumulatively across many histogram bins. GLCM statistics and vein features received comparatively lower scores, indicating that while these modalities provide complementary discriminative information, their marginal contribution is smaller relative to color and shape. This finding suggests that future work could prioritize richer color space decompositions and finer-grained color histogram representations, while treating texture and vein features as supplementary rather than primary descriptors.

6.4 Misclassification Patterns

Inspection of the confusion matrix for the SVM with Top-N Feature Selection reveals that the majority of misclassifications involve species pairs with high morphological similarity. Notably, species within the same botanical family tend

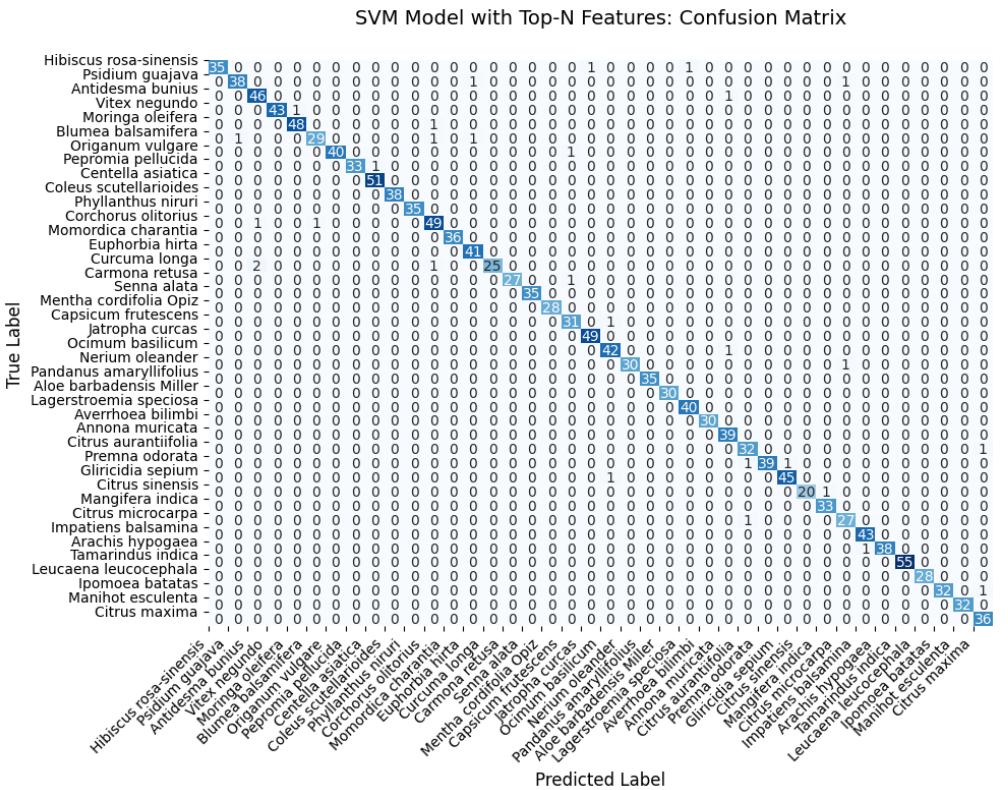
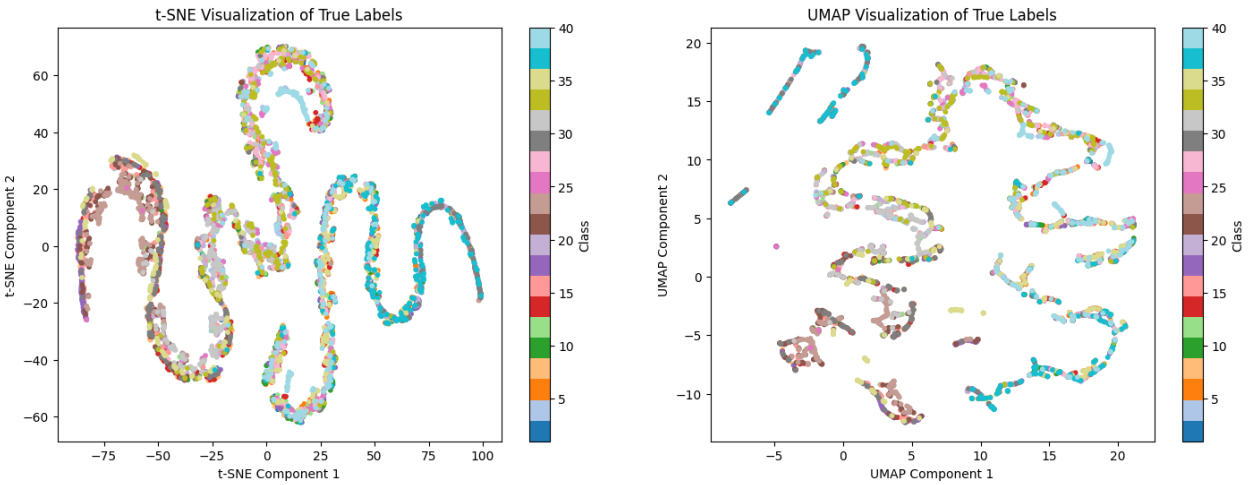


Figure 3: Confusion matrix of the SVM with Top-N Feature Selection on the 40-class test set.

to exhibit higher rates of mutual confusion. Species with atypical morphologies relative to the majority of the dataset, such as compound-leaved species and species with highly irregular margins, showed lower per-class recall, suggesting that the feature extraction pipeline captures the dominant leaf morphology more reliably than it captures structural outliers. These observations are consistent with the pattern reported in analogous multi-class leaf classification studies [9] and suggest that targeted augmentation of training samples for morphologically atypical classes, combined with additional vein and margin descriptors, could reduce the residual misclassification rate.

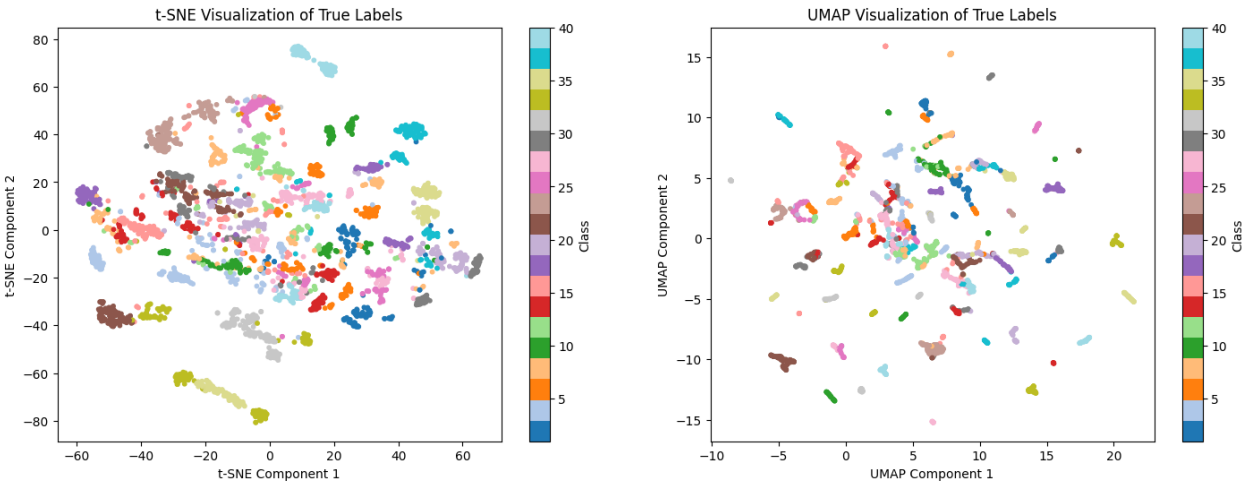
6.5 Practical Implications

The results of this study carry practical implications for the deployment of plant identification systems in Philippine healthcare and conservation contexts. The SVM pipeline with Top-N Feature Selection, which achieves 98.06% accuracy on a 40-class dataset using a compact subset of 200 hand-crafted features, is substantially more lightweight than a VGG19-based system. SVM inference on a 200-dimensional feature vector requires negligible computation relative to a forward pass through a 19-layer convolutional network, making the proposed pipeline well-suited to deployment on resource-constrained hardware, including mobile devices used by rural health practitioners, field botanists, and community herbalists who may lack reliable access to cloud-based inference infrastructure. Furthermore, the interpretability of the hand-crafted features and the ANOVA F-score ranking provide a transparent account of the model's decision-making process, which is a meaningful practical advantage in domains where model explainability is valued.



(a) t-SNE (Raw)

(b) UMAP (Raw)



(c) t-SNE (Top-N)

(d) UMAP (Top-N)

Figure 5: t-SNE and UMAP embedding spaces generated before (raw) and after Top-N feature selection.

7. CONCLUSION

This study investigated whether a classical machine learning pipeline, combining hand-crafted image feature extraction, Top-N feature selection, and Grid Search-optimized SVM classification, can achieve classification accuracy competitive with deep learning benchmarks on a 40-class Philippine medicinal plant leaf dataset.

The experimental results demonstrate that the optimized SVM with Top-N Feature Selection achieved 98.06% test accuracy, surpassing the VGG19 + CNN benchmark of 92.67% by 5.39 percentage points and falling within 0.33 percentage points of the VGG19 Feature Extractor + SVM baseline. These results confirm that traditional feature extraction and classical machine learning methods can achieve results comparable to deep learning approaches in plant classification tasks.

ANOVA F-score analysis confirmed color as the dominant discriminative modality, with shape features providing secondary contributions and texture and vein features playing supplementary roles.

Several directions for future work are identified. First, the dataset could be expanded to include additional Philippine medicinal plant species, a broader range of imaging conditions such as varying lighting and background environments, and images captured from multiple leaf surfaces and developmental stages, which would test the generalization of the pipeline beyond the controlled conditions of the current dataset. Second, the feature extraction stage could be enriched with additional color space decompositions and finer-grained color histogram representations, which may reduce the residual misclassification rate among morphologically similar species. Finally, field validation of the proposed pipeline on a mobile deployment platform would provide direct evidence of its practical utility for rural health practitioners and field botanists operating in resource-constrained environments across the Philippine archipelago.

8. REFERENCES

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